Rahul Lohra

+91-9771315217 | rahullohra987@gmail.com | linkedin.com/in/rahul-lohra | github.com/rahullo | portfolio

Professional Summary

Results-driven Software Engineer with hands-on experience in full-stack development, microservices architecture, and secure application development. Proficient in Python, JavaScript, and Java with expertise in Spring Boot, Django, and React.js. Demonstrated ability to design and implement RESTful APIs, maintain high test coverage, and deliver scalable solutions in Agile environments. Strong background in security implementation, CI/CD pipelines, and cross-functional collaboration with focus on code quality and performance optimization.

Technical Skills

Programming Languages: Python (Advanced), JavaScript (Advanced), Java, SQL, PL/SQL, HTML/CSS
Backend Frameworks: Spring, Spring Boot, Django, Flask, Node.js, Express.js, Microservices Architecture
Frontend Technologies: React.js (Hooks, Context API, Redux), Angular, HTML5, CSS3, Responsive Design
Database & ORM: PostgreSQL, MongoDB, MySQL, RDBMS, JPA, Hibernate, JDBC, Database Design & Optimization
DevOps & Cloud: Jenkins CI/CD, Docker, Git/GitHub, Tomcat, AWS, Azure, GCP, Pivotal Cloud Foundry
API Development: REST API Design, API Integration, Third-party Services (Twilio, Gmail), Documentation
Testing & Security: Unit Testing, JUnit, Pytest, Test Automation, PBKDF2, OAuth, Multi-factor Authentication
Tools & Methodologies: Agile/Scrum, Jira, Sonar, Object-Oriented Design, Design Patterns, Code Reviews

Education

Banaras Hindu University (BHU) — Varanasi, India
Master of Computer Applications (MCA) – CGPA: 7.3/10 — *Sept 2023 – June 2025*

Dr. Shyama Prasad Mukherjee University — Ranchi, India
Bachelor of Technology in Information Technology – CGPA: 8.61/10 — *June 2020 – Aug 2023*

Professional Experience

Software Engineering Intern — Feb 2025 – Aug 2025
CERT-In (Indian Computer Emergency Response Team) — *Delhi, India*

- Architected and deployed enterprise-grade full-stack web application using Django framework serving 100+ internal users with role-based access control and automated document processing, improving organizational workflow efficiency by 40%
- Designed and implemented RESTful APIs following microservices architecture principles for authentication, document management, and notification services with proper error handling and request validation, ensuring 99% uptime
- Developed secure backend application logic with PBKDF2-SHA256 password hashing, multi-factor authentication using OTP validation, and Django Axes for brute-force protection, achieving zero security incidents and maintaining compliance
- Integrated third-party services including Twilio API for SMS notifications and Gmail API for automated email workflows, reducing manual verification time by 60% and improving user authentication experience
- Maintained 90%+ unit test coverage using pytest framework, implemented automated testing in Jenkins CI/CD pipeline, and ensured all new code maintained or exceeded current coverage standards with comprehensive test suites

Technical Projects

AI-Powered File Converter & Document Summarizer | *React.js, Django, Flask, REST APIs, NLP* Nov 2024 – Feb 202

- Built scalable document processing platform with React.js frontend and Django/Flask backend, enabling users to convert multiple file formats (PDF, DOCX, TXT) and generate AI-powered summaries with 90%+ accuracy
- Engineered robust REST API endpoints for asynchronous file processing with comprehensive error handling, input validation, and rate limiting to ensure 99% uptime and seamless user experience across concurrent requests
- Developed responsive single-page application using modern React patterns including Hooks, Context API for state management, and implemented real-time progress tracking for long-running summarization tasks
- Integrated state-of-the-art NLP models for intelligent text extraction and summarization, implementing caching mechanisms that reduced processing time by 5x compared to manual review while maintaining high accuracy

Federated Learning System for Healthcare Data | *Python, Scikit-learn, Machine Learning* Mar 2024 – Aug 2024

- Designed and implemented privacy-preserving Federated Learning framework for sensitive medical insurance dataset across 5 distributed clients, demonstrating secure decentralized model training without raw data sharing

- Developed modular Python-based machine learning pipeline using SGDRegressor with comprehensive data preprocessing including normalization, feature engineering, and outlier detection achieving 85% prediction accuracy
- Implemented custom weight aggregation algorithm for federated averaging, handled client synchronization, and ensured model convergence across distributed nodes while maintaining data privacy standards

Core Competencies & Soft Skills

Technical Competencies: Full-Stack Development, Microservices Design, API Architecture, Database Optimization, Security Implementation, Performance Tuning, Cloud Deployment, System Design, Technical Documentation
Soft Skills: Strong Verbal & Written Communication, Quick Learning & Adaptability, Problem-Solving, Attention to Detail, Team Collaboration, Time Management, Ability to Handle Multiple Concurrent Activities